Basic Principles of Medicine 2
Module: Digestion and Metabolism

DM 04: Gluconeogenesis

Dr. Clayton Taylor, M.D.

ctaylor2@sgu.edu

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Acknowledgements:
Dr. Felicia Ikolo
Dr. Margit Trotz

Recommended Reading

- Lippincott's Biochemistry 9th Edition

Gluconeogenesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library

- 🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402

Objectives: Gluconeogenesis

SOM. MK.I.BPM2.2.DM.3.BCHM.1132	Describe the purpose of gluconeogenesis and indicate the organs and cellular compartments where gluconeogenesis occurs.
SOM. MK.I.BPM2.2.DM.3.BCHM.1133	Outline hepatic gluconeogenesis (irreversible steps, substrates).
SOM. MK.I.BPM2.2.DM.3.BCHM.1134	Justify why there can be no net production of glucose from the carbons of acetyl CoA or from fatty acid degradation leading to acetyl CoA.
SOM. MK.I.BPM2.2.DM.3.BCHM.1135	Discuss the glucose-alanine cycle and the Cori cycle and the formation of pyruvate in the liver.
SOM. MK.I.BPM2.2.DM.3.BCHM.1136	Describe the reaction catalyzed by pyruvate carboxylase (biotin and acetyl CoA as activator).
SOM. MK.I.BPM2.2.DM.3.BCHM.1137	Discuss the reaction catalyzed by PEP carboxykinase (GTP) and glucose 6-phosphatase.
SOM. MK.I.BPM2.2.DM.3.BCHM.1138	Describe how precursors (glucogenic amino acids, lactate and glycerol) are channeled into gluconeogenesis during starvation.
SOM. MK.I.BPM2.2.DM.3.BCHM.1139	Describe the reaction catalyzed by fructose 1,6-bisphosphatase and its inhibition by AMP and F 2,6-bisP.
SOM. MK.I.BPM2.2.DM.3.BCHM.1140	Describe the hepatic bifunctional enzyme and distinguish the role of glucagon and insulin on gluconeogenesis.
SOM. MK.I.BPM2.2.DM.3.BCHM.1141	Explain the effect of glucose 6-phosphatase deficiency in both gluconeogenesis and glycogen degradation.
SOM. MK.I.BPM2.2.DM.3.BCHM.1142	Discuss the importance of beta-oxidation for provision of ATP (energy) and allosteric regulators for gluconeogenesis
SOM. MK.I.BPM2.2.DM.3.BCHM.1143	Outline the role of fatty acid oxidation in providing ATP for gluconeogenesis in liver and highlight the role of acetyl CoA as an allosteric regulator of gluconeogenesis.

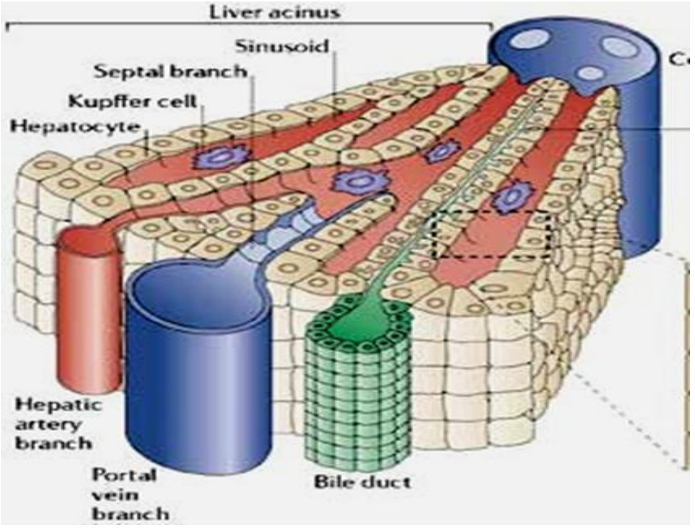
Tips for covering the biochemistry and genetic components of medical school

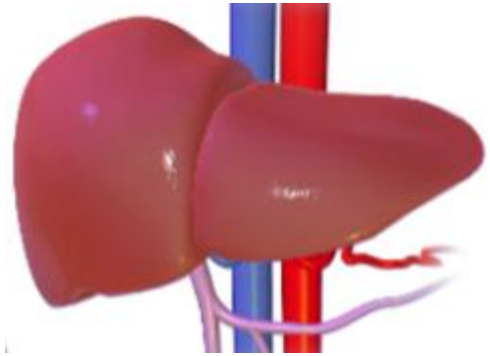
- Genetics: **Definitions**, population genetics
- Biochemistry: **Pathways**, **Regulation of pathways, disorders,** and molecular **structure-function** relationships.
- **Clinical and pharmacological correlations**

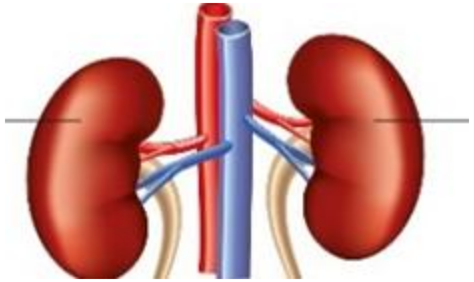
Gluconeogenesis

- Glucose synthesized from **NON-CARBOHYDRATE** precursors.
- Occurs in:
  - Liver (periportal hepatocytes)
  - Renal cortex cells

Tissues that contain **Glucose-6-Phosphatase**







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Importance of Gluconeogenesis

- During fasting the liver maintains blood glucose levels via
  - Glycogenolysis (active up to 24 hours) and
  - **Gluconeogenesis (active in overnight fast and only source of glucose after 1 day of fasting)**
- Glucose main fuel for: Brain, red blood cells, kidney medulla, lens and cornea of the eye, testes and exercising muscle

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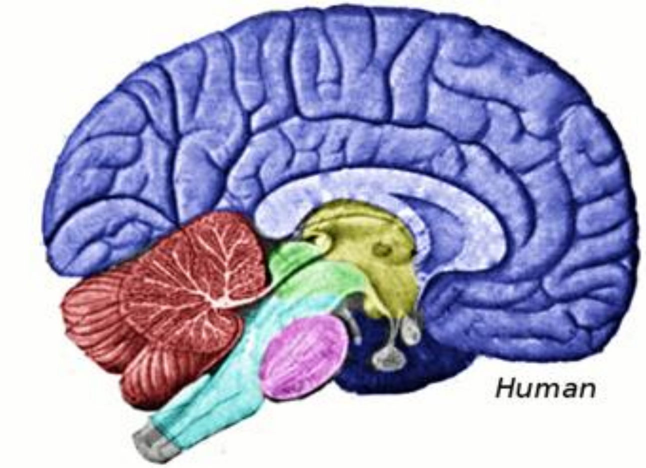
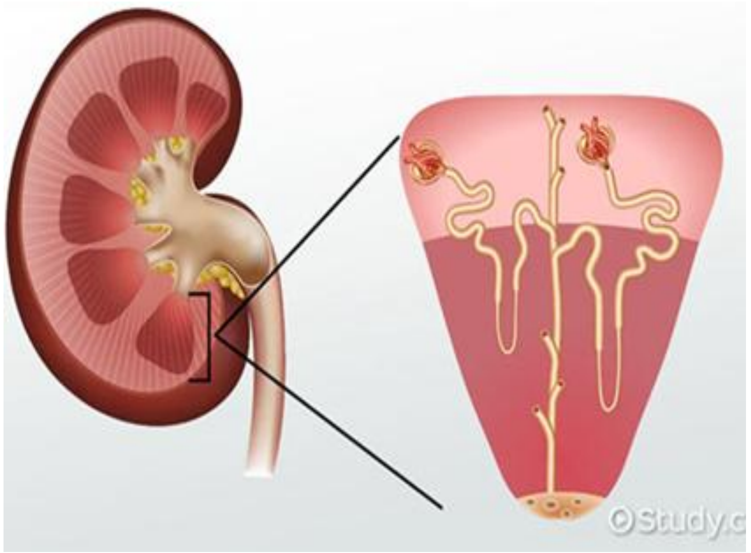
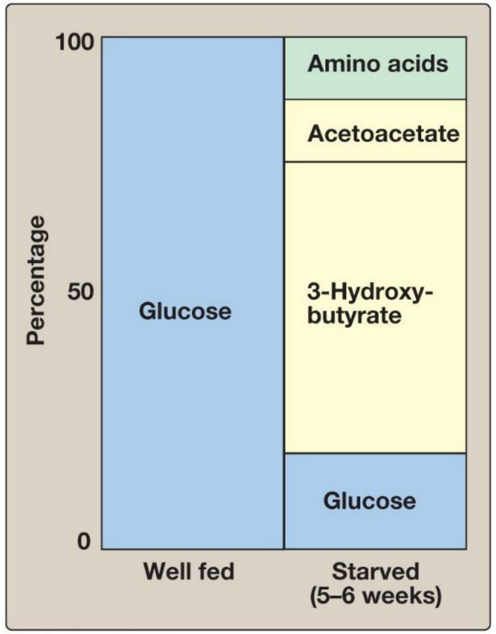
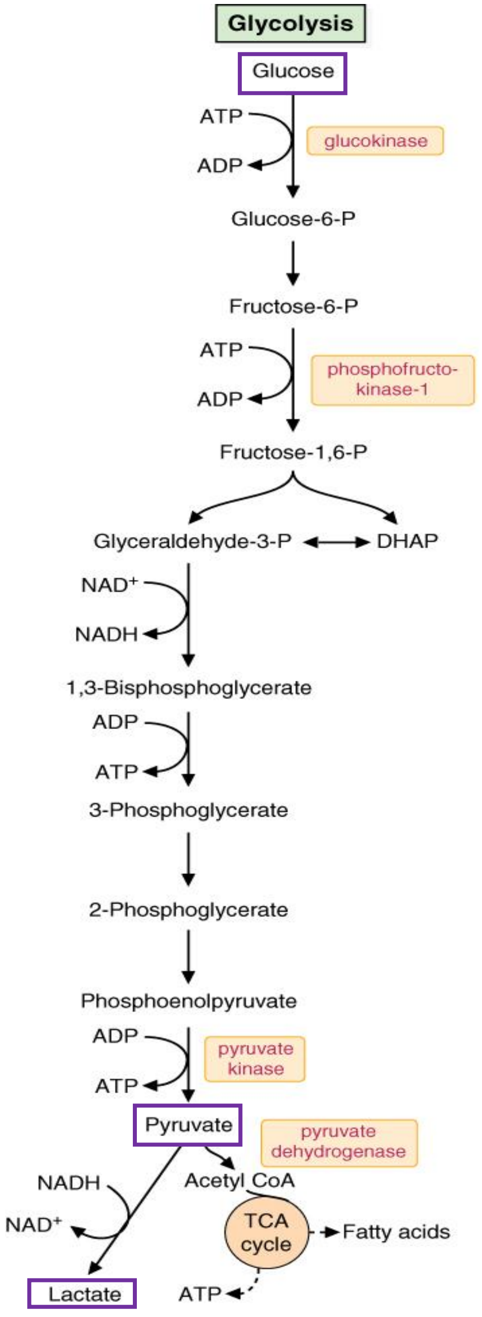


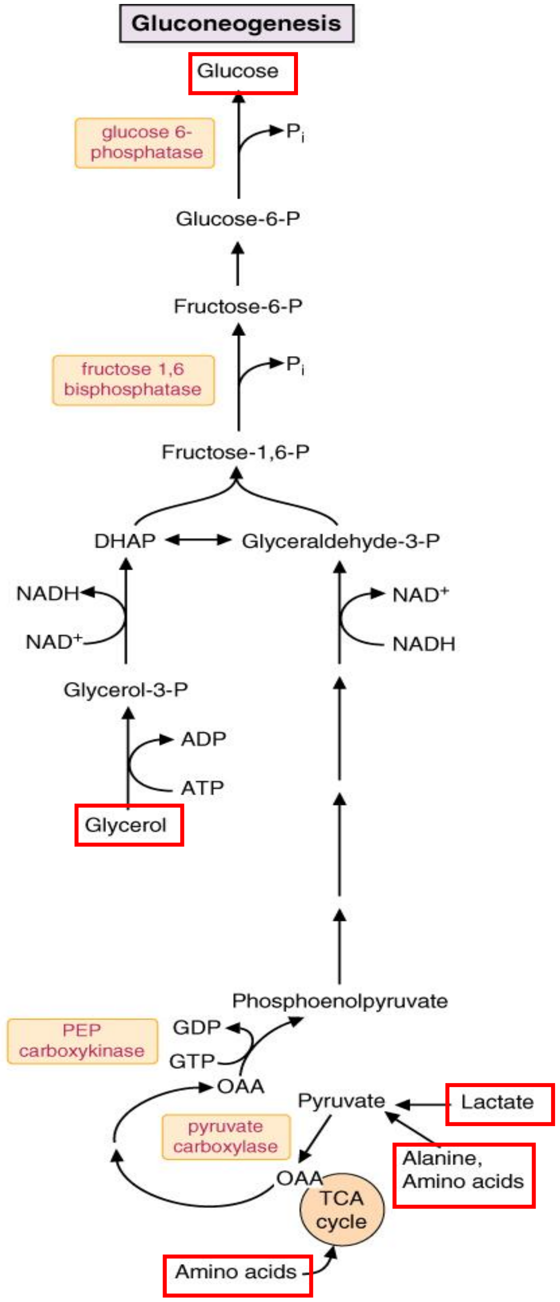


Importance of Gluconeogenesis

- Sole provider of glucose when hepatic glycogen depleted (after 1 day fasting).
- During periods of low blood glucose levels: fasting, starvation and during the flight or fight response.
- Activated by Glucagon and Epinephrine





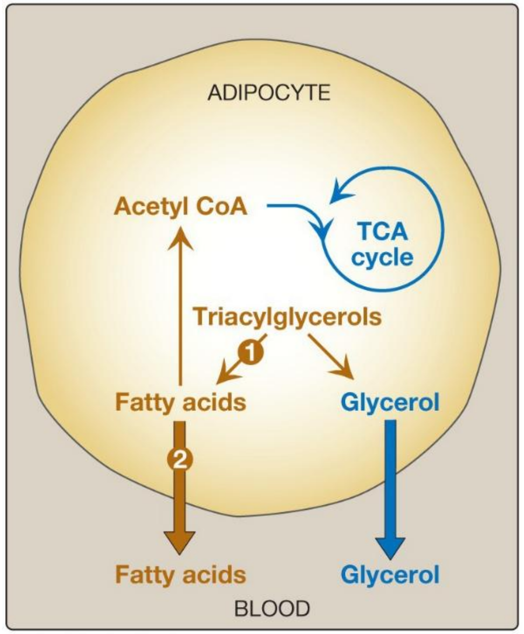


Gluconeogenesis vs Glycolysis

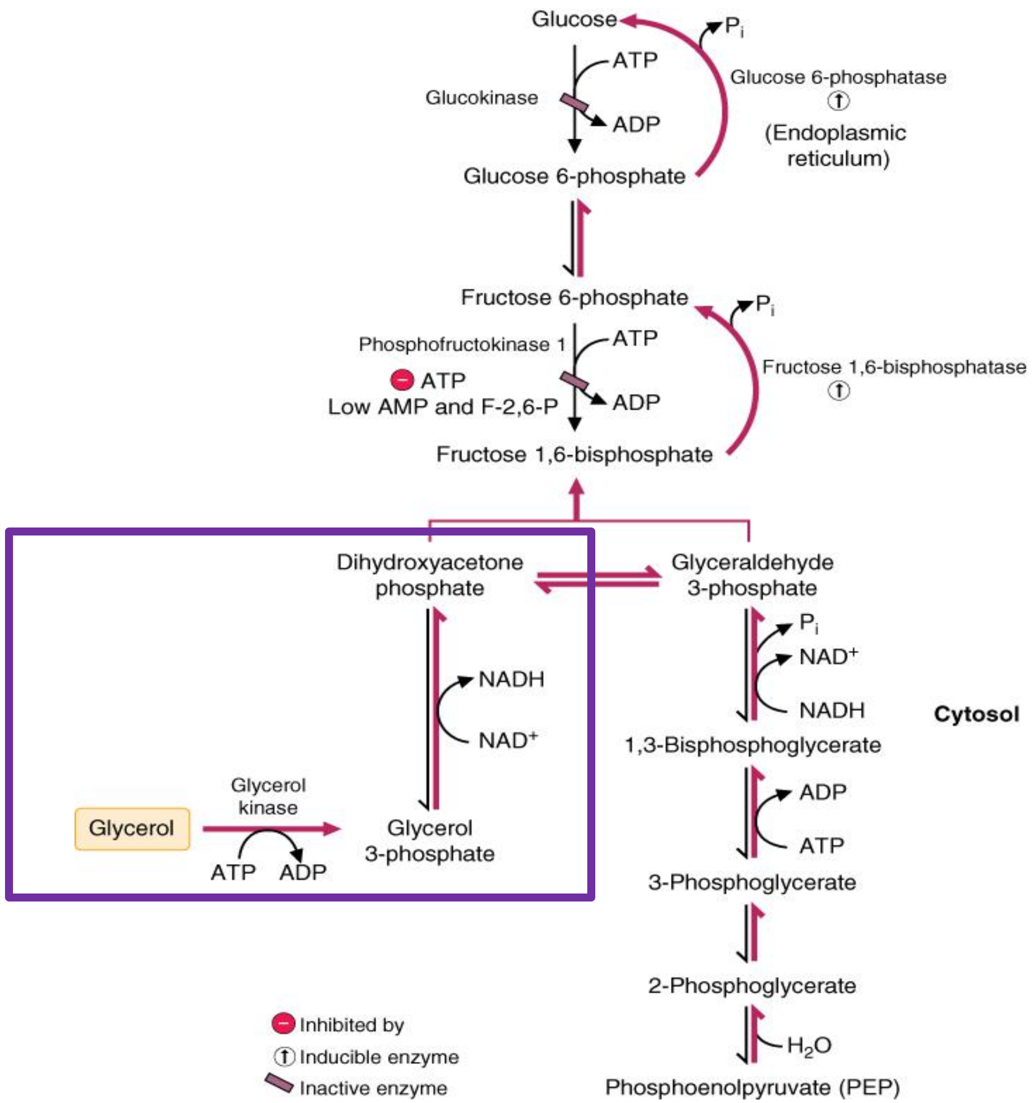
- Gluconeogenesis almost reverse of glycolysis: except for three irreversible glycolysis reactions
- **Four irreversible reactions** in gluconeogenesis
- All other steps are reversible
- Only one pathway can be active while the other is inhibited (under a specific metabolic condition) → **Reciprocal regulation**

Substrates of Gluconeogenesis

- **Lactate** (from anaerobic glycolysis in peripheral tissues; Cori cycle)
- **Glucogenic amino acids** especially **Alanine** (muscle breakdown)
- **Glycerol** (from TAG breakdown from adipose tissue)



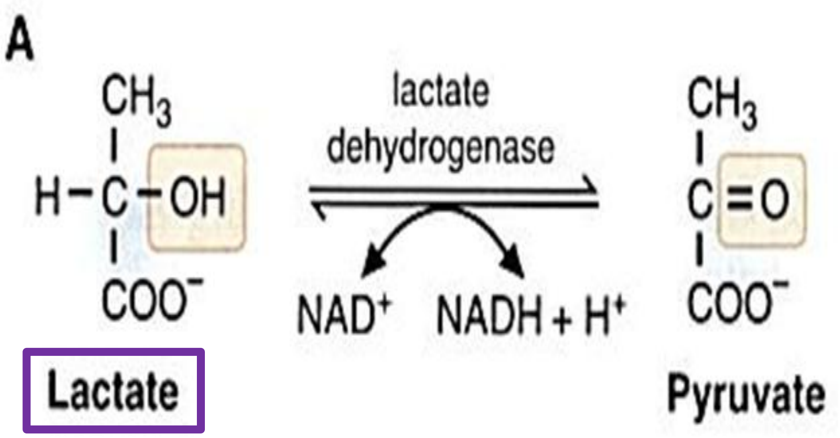
Substrates of Gluconeogenesis: Glycerol



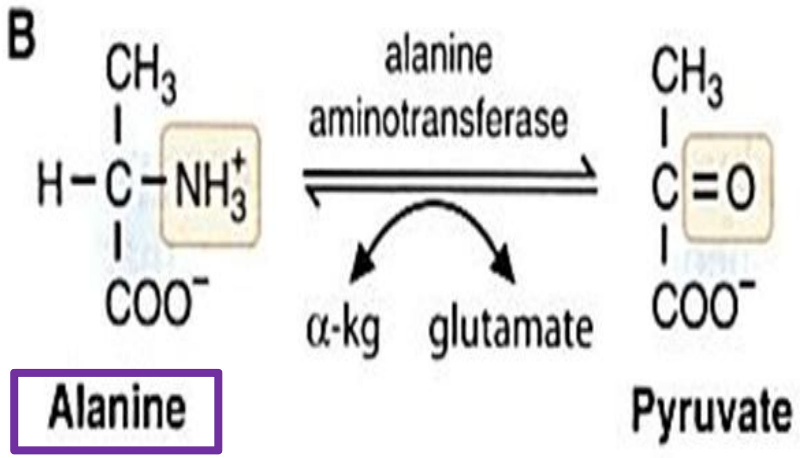
- Triglycerides (TAG) are broken down during fasting to form glycerol and free fatty acids (which undergo β oxidation)
- **Glycerol** forms dihydroxyacetone phosphate (DHAP), a glycolytic intermediate → Enters gluconeogenesis

Substrates of Gluconeogenesis: Lactate and Alanine

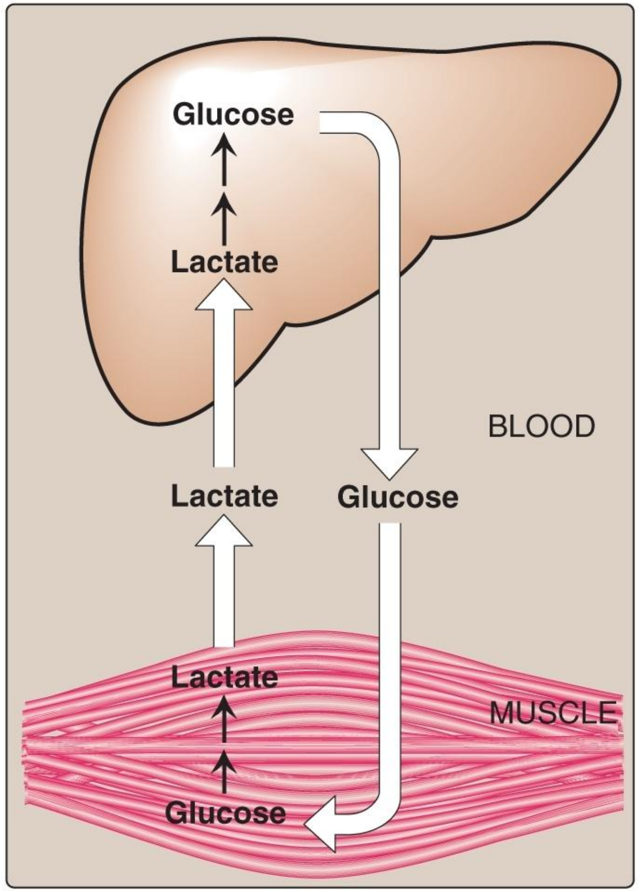
Lactate is oxidized to Pyruvate by Lactate dehydrogenase (LDH).



Alanine is transaminated to Pyruvate by Alanine aminotransferase (ALT).



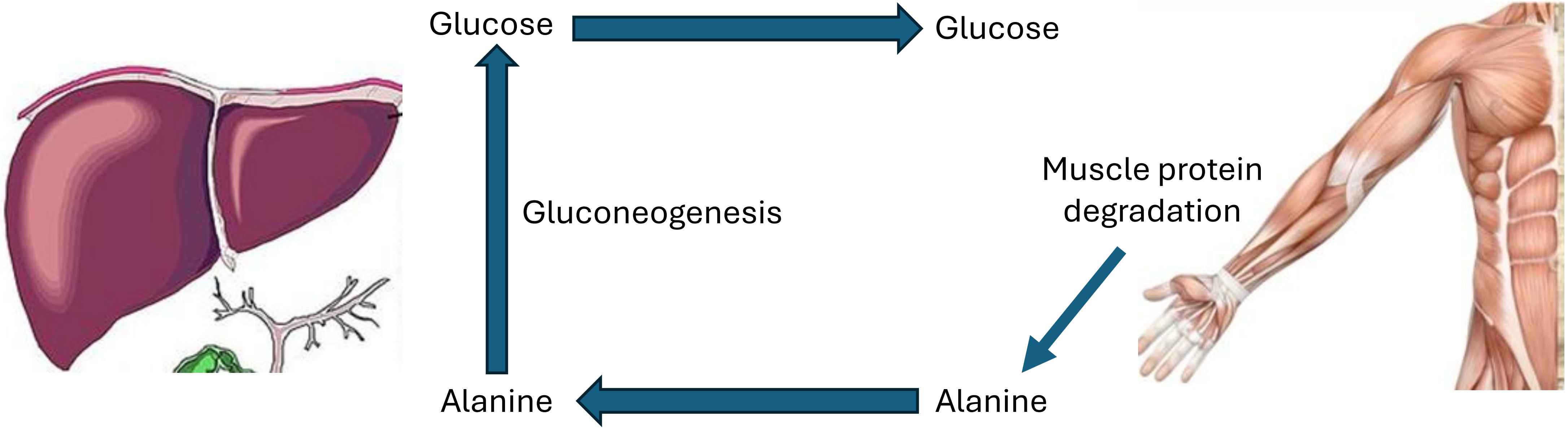
Cori Cycle (Glucose-lactate cycle)



- Lactate produced by anaerobic glycolysis in skeletal muscle and red blood cells is transported to the liver and converted to glucose via gluconeogenesis

Glucose-Alanine Cycle

- Alanine from muscle transported to liver for gluconeogenesis
- Active during fasting



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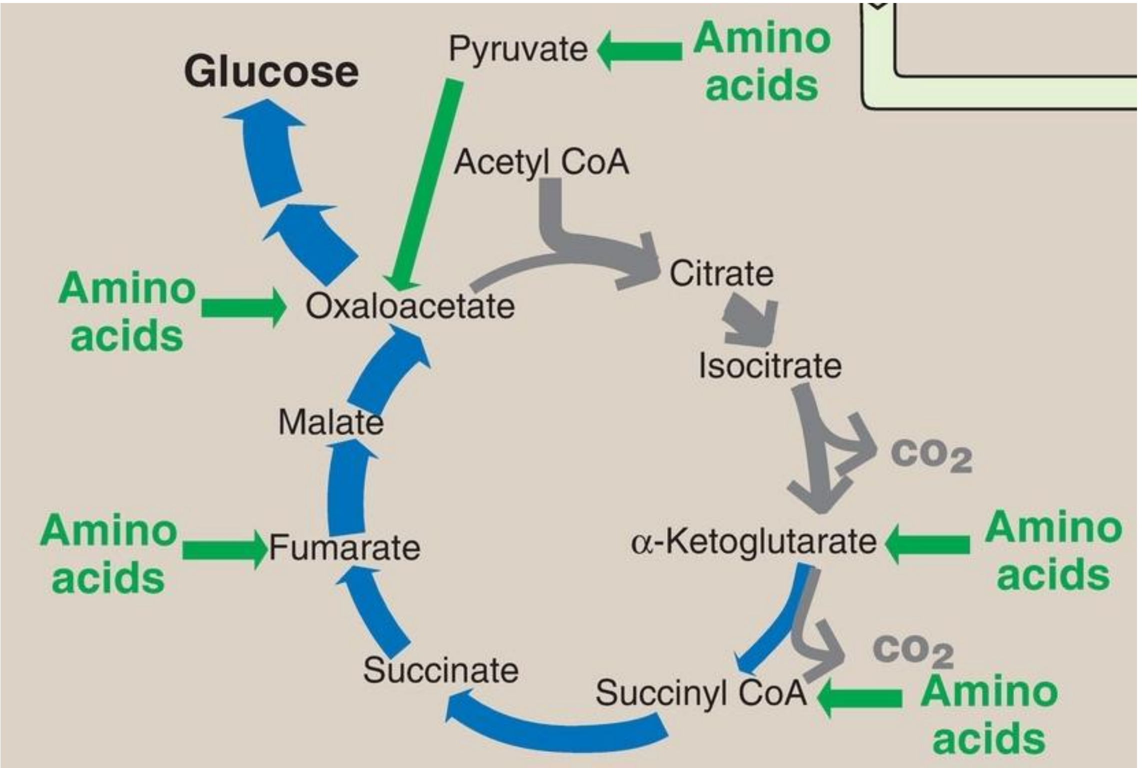
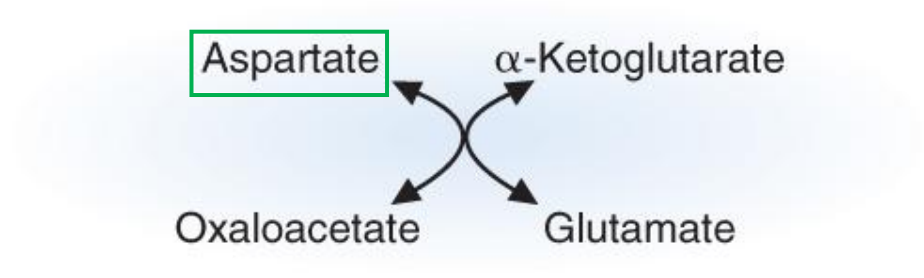
Substrates of Gluconeogenesis: **Glucogenic amino acids**





	Glucogenic	Ketogenic	Glucogenic and Ketogenic
	Alanine Asparagine Aspartate Cysteine Glutamate Glutamine Glycine Proline Serine Arginine* Threonine Histidine Methionine Valine	Leucine Lysine	Tyrosine Isoleucine Phenylalanine Tryptophan

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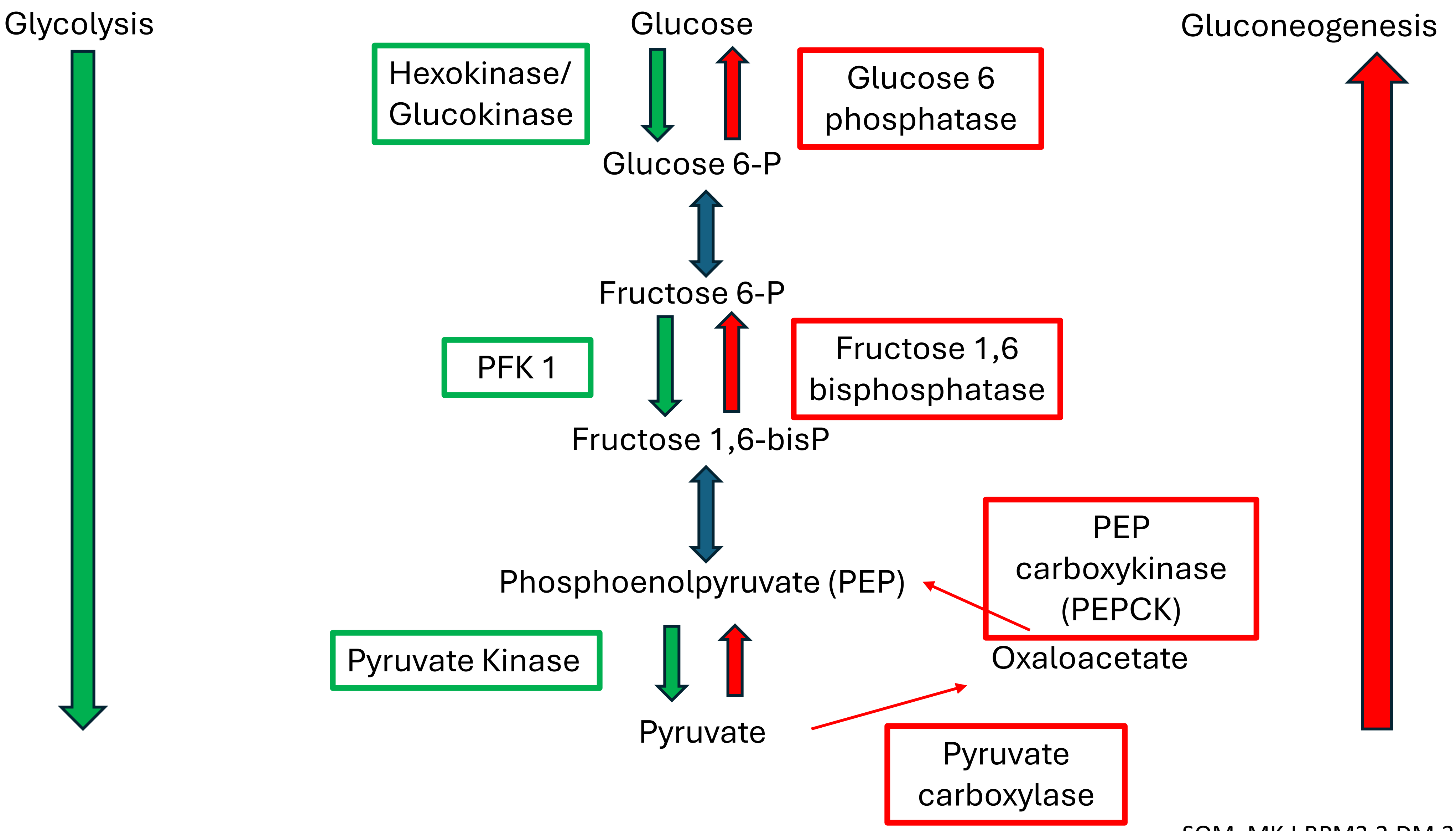
Gluconeogenesis vs Glycolysis (Enzymes)



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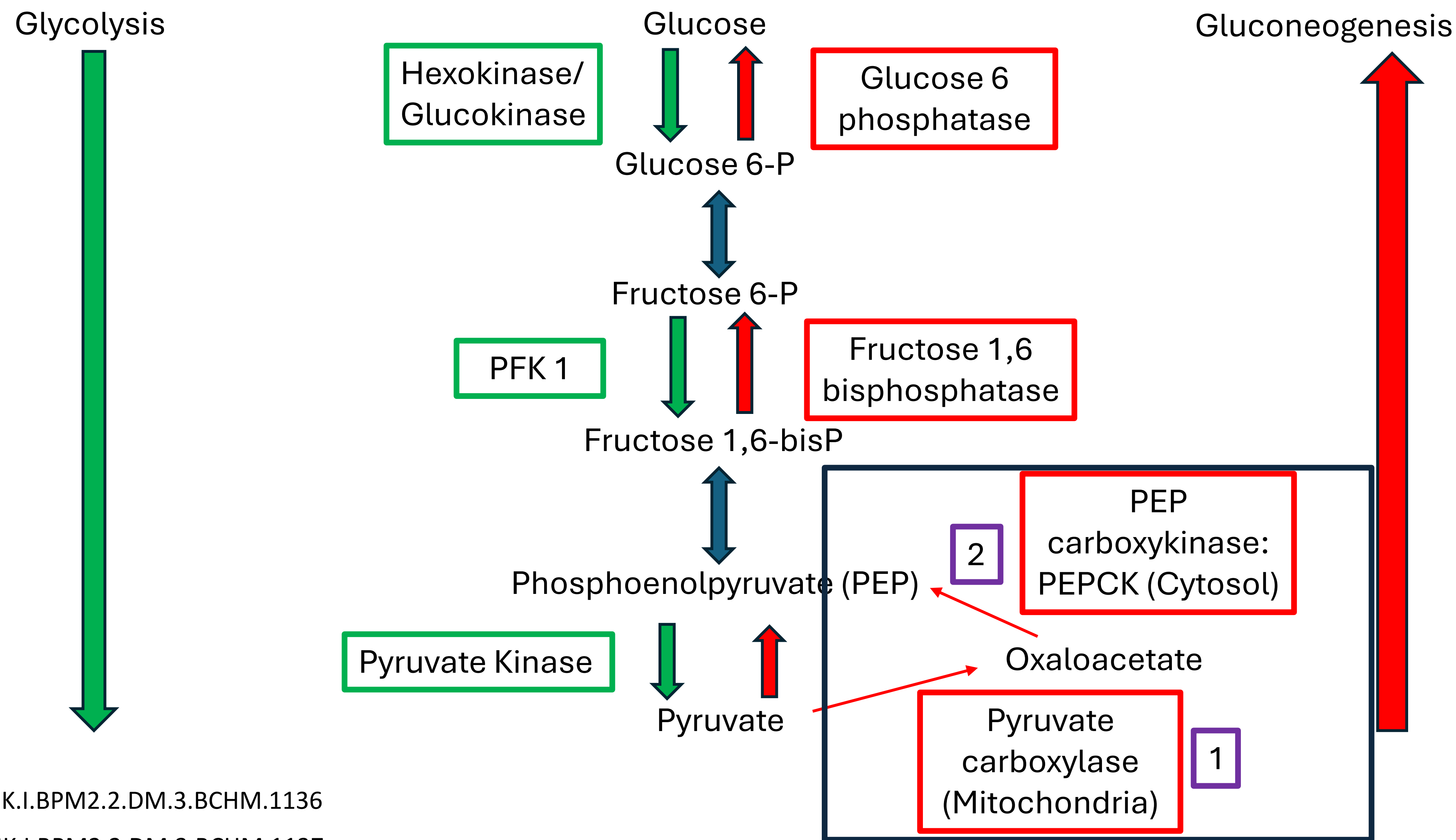
Pyruvate to Phosphoenolpyruvate



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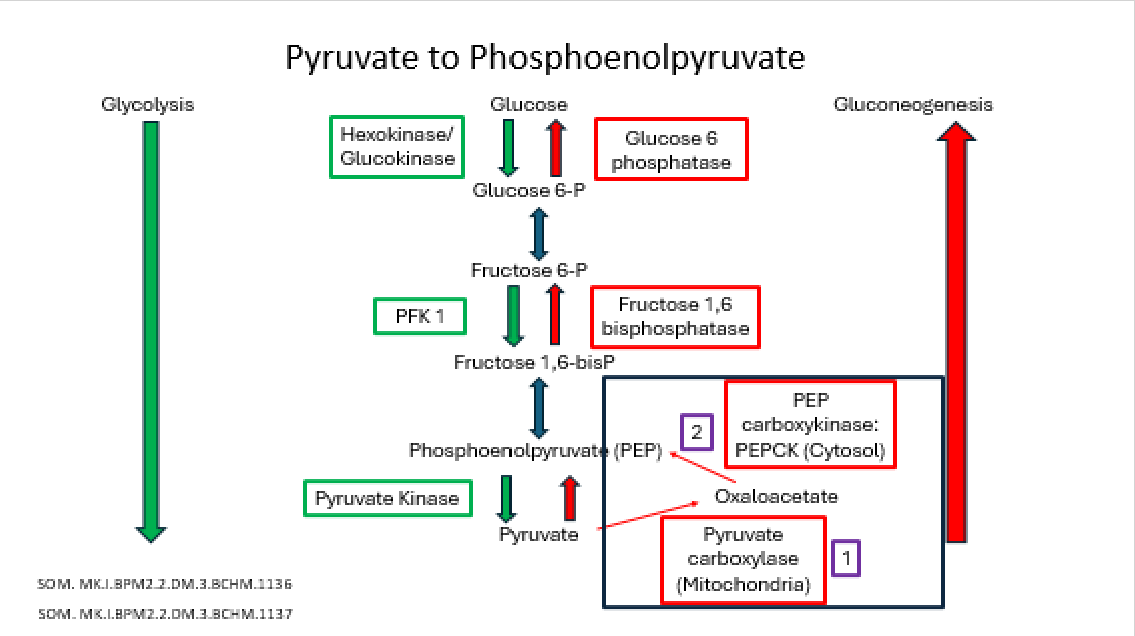
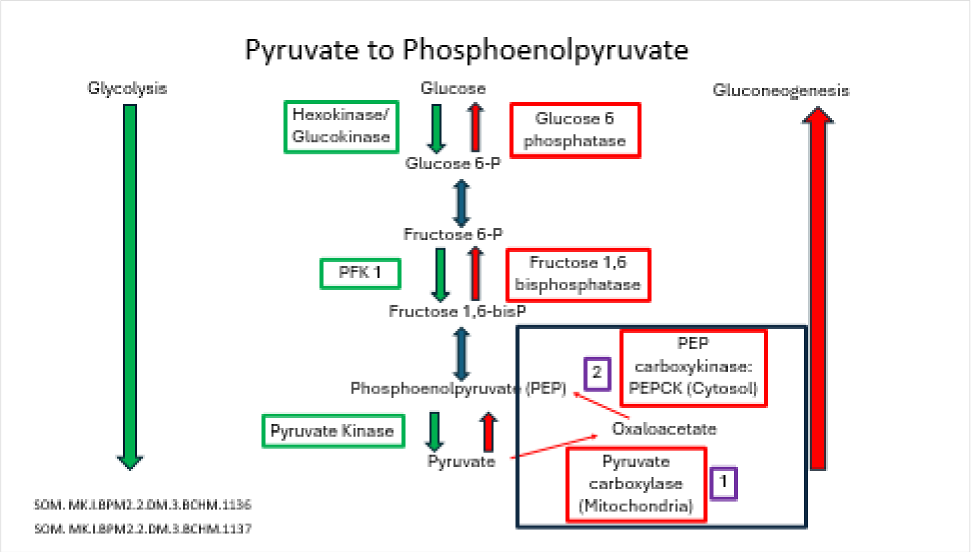
Step 1: Pyruvate carboxylase: Pyruvate to Oxaloacetate



- Mitochondria
- Pyruvate → Oxaloacetate
- Needs **ABC**: **ATP, Biotin and carbon dioxide**
- **Biotin (vitamin B7) coenzyme**
- Acetyl CoA (from fatty acid β-oxidation): **Absolute allosteric activator.**
- Oxaloacetate (cannot leave mitochondria) → Malate

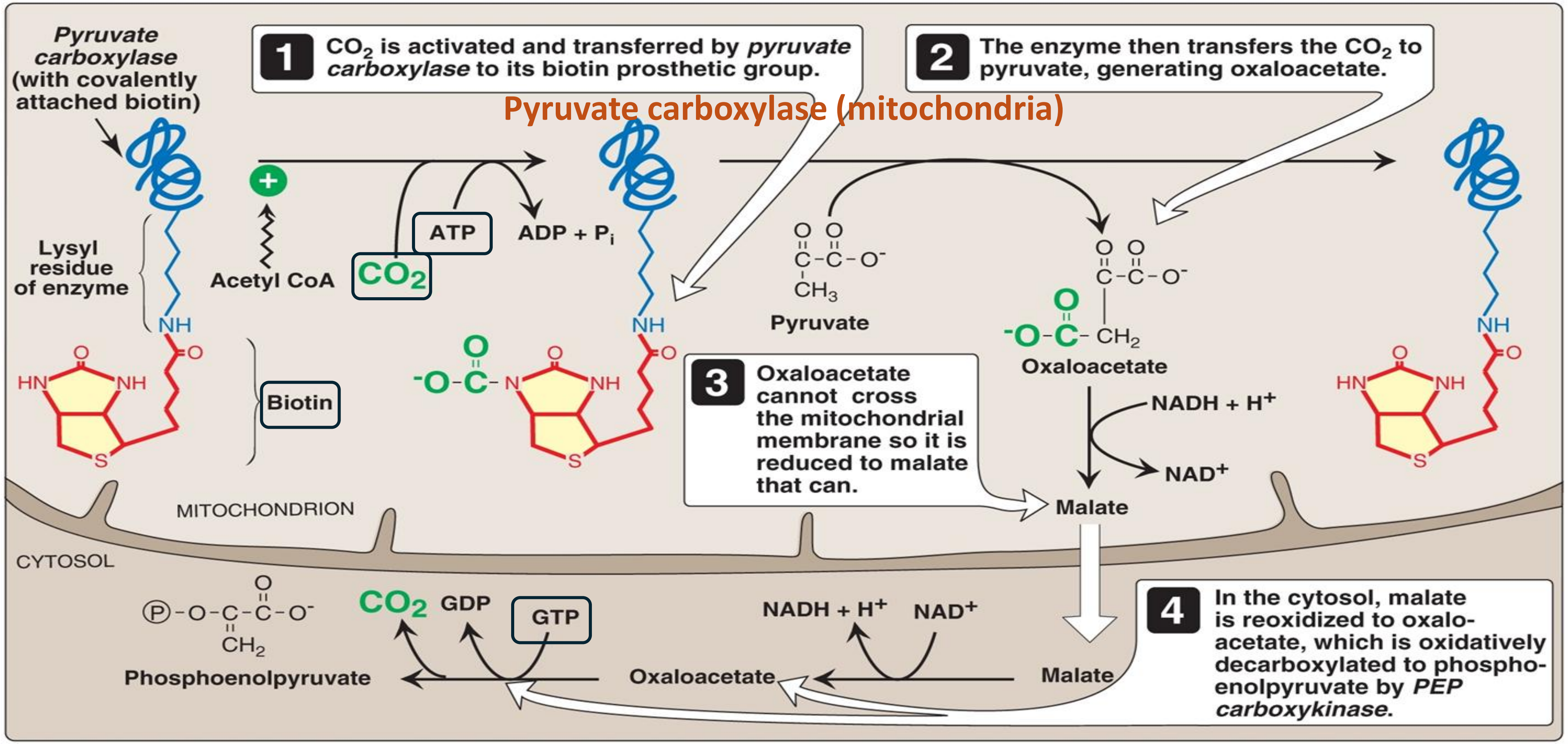
Step 2: Phosphoenolpyruvate Carboxykinase (PEPCK): Oxaloacetate to Phosphoenolpyruvate



- Cytosol
- In the cytosol: Malate → Oxaloacetate → Phosphoenolpyruvate
- PEPCK uses GTP (not ATP); Releases CO_2
- Induced (increased gene transcription) by
  - Long term regulation
  - Cortisol → High plasma glucose in Cushing syndrome
  - Glucagon (fasting)
  - Epinephrine (stress)

Conversion of Pyruvate to phosphoenolpyruvate

Step 1 and 2: Pyruvate carboxylase (mitochondria); PEPCK (cytosol)



PEPCK (cytosol)

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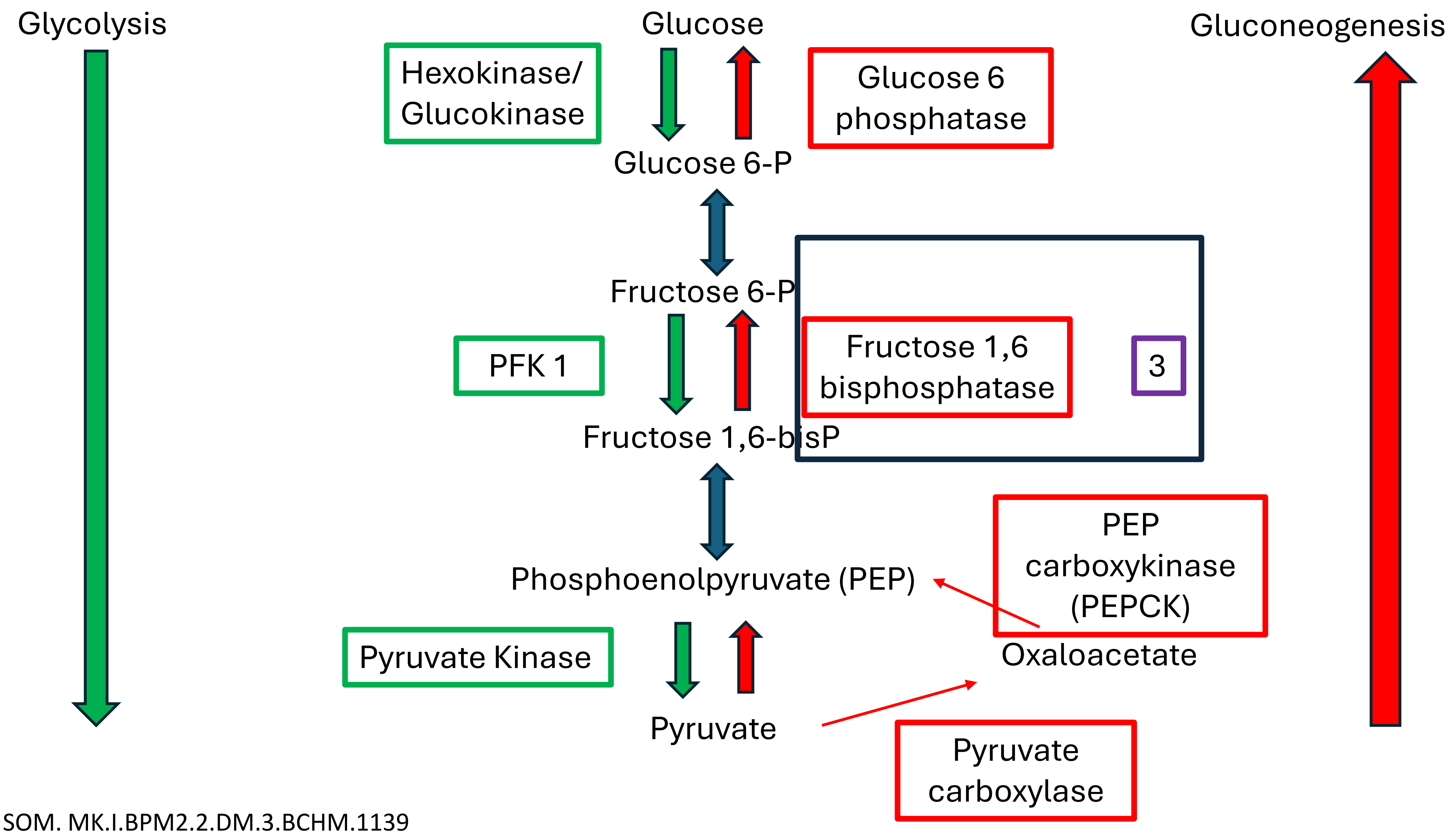
Step 3: Fructose 1,6-bisphosphatase



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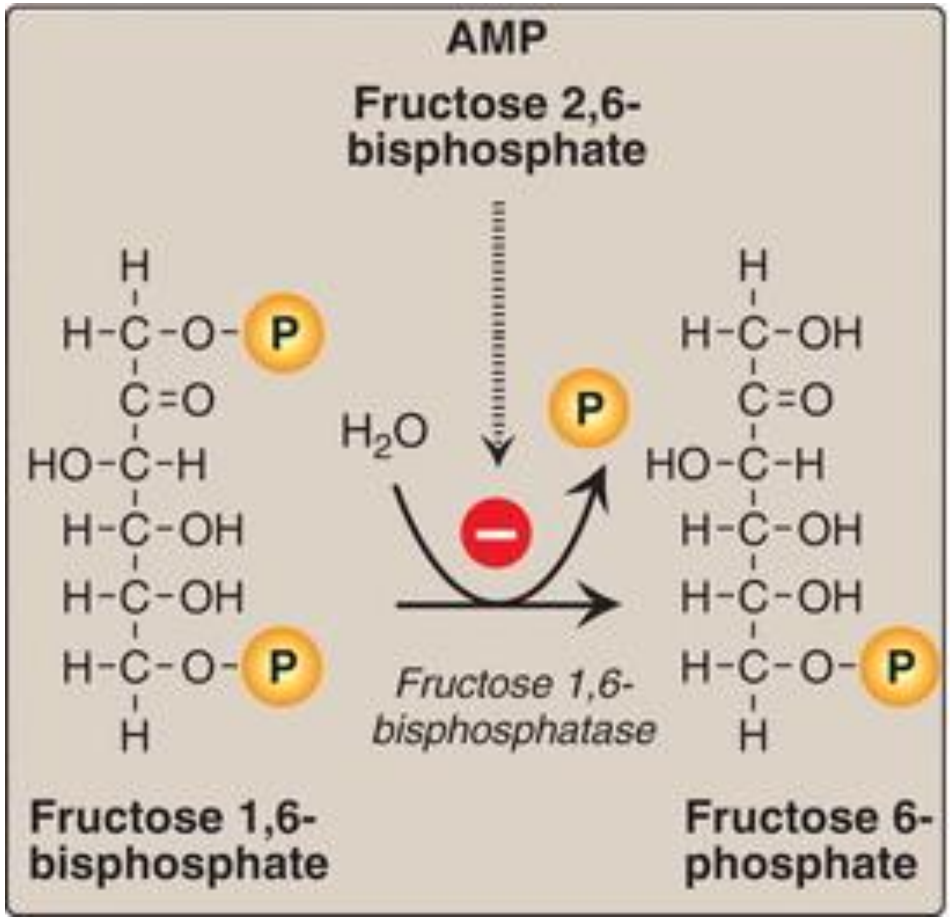
Step 3: Fructose 1,6 Bisphosphatase: Fructose 1,6 bisphosphate to Fructose 6-phosphate



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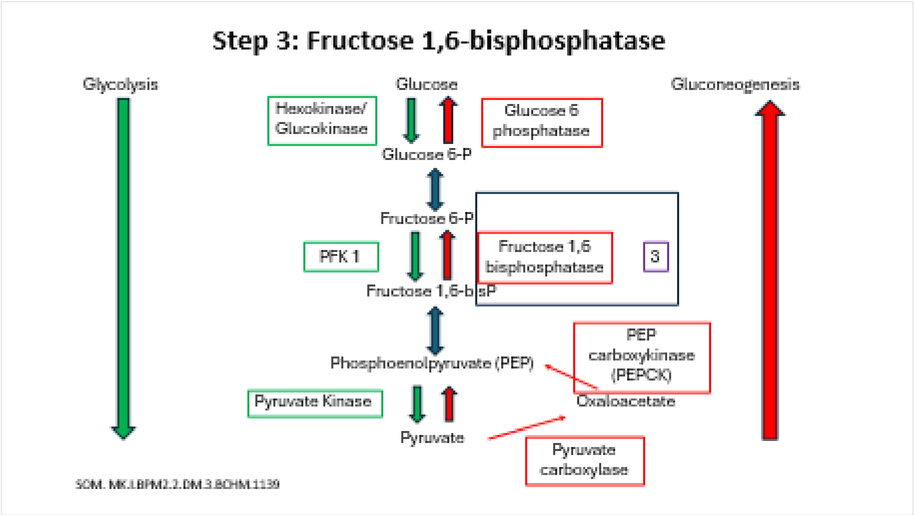
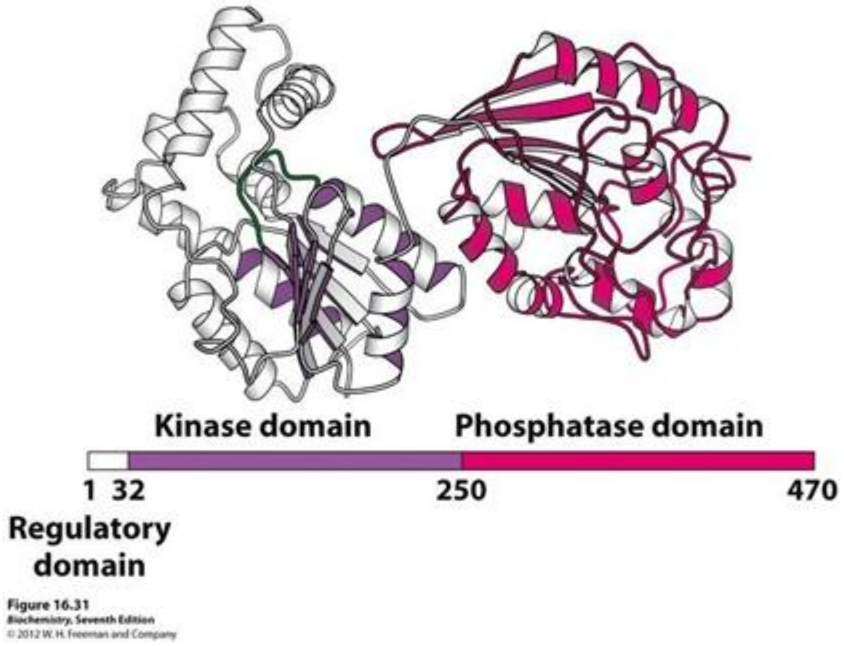
- Cytosol
- **Regulated enzyme** of gluconeogenesis
- Allosterically inhibited by **Fructose 2,6-bisphosphate** and AMP (low energy status)



Allosteric Regulation:

Fructose 2,6 bisphosphate inhibits gluconeogenesis and stimulates glycolysis (Allosteric regulation)

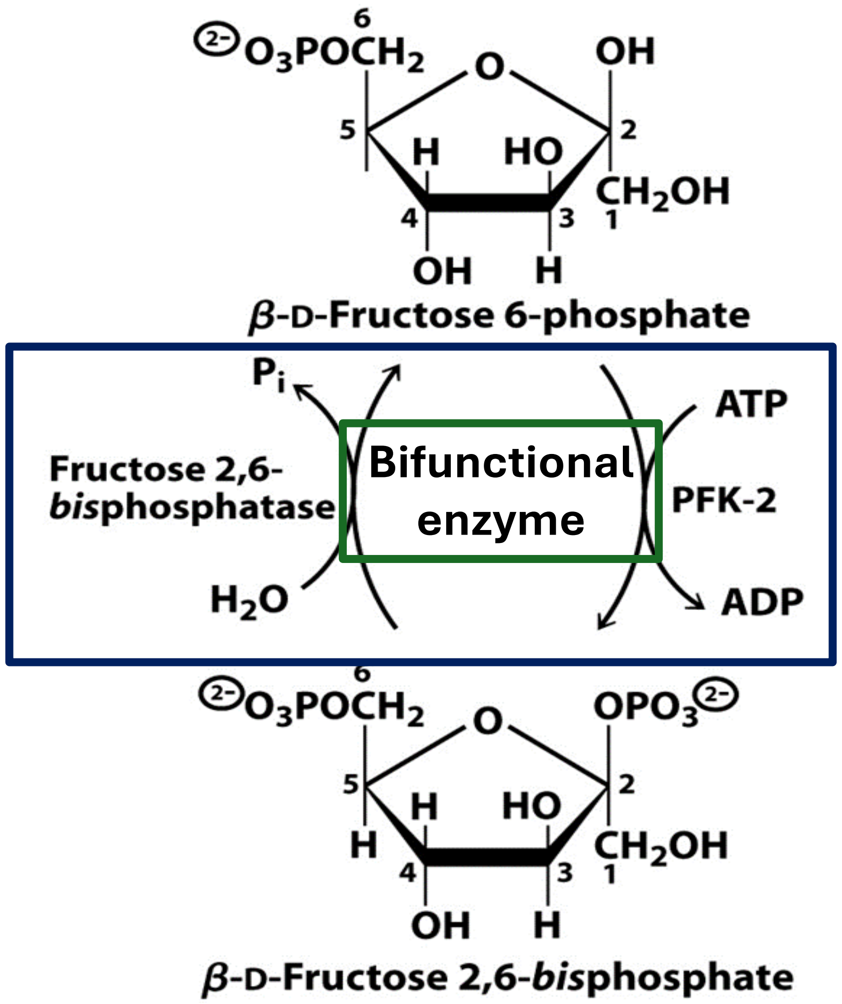


- **Bifunctional Enzyme** (PFK-2 and FBPase-2)
  - Has 2 domains (PFK-2 and FBPase-2)
  - Forms F2,6-bisP by PFK-2 activity and
  - Degrades F2,6-bisP by FBPase-2 activity
  - **Only one activity can be performed at any time, either PFK-2 or FBPase-2.**

- **Fructose 2,6-bisphosphate (F2,6-bisP)** formed by bifunctional enzyme during the fed/postprandial phase → **Inhibits gluconeogenesis and stimulates glycolysis.**

- **The opposite occurs during fasting**

Hormone Regulation: of Glycolysis and Gluconeogenesis (Bifunctional Enzyme)



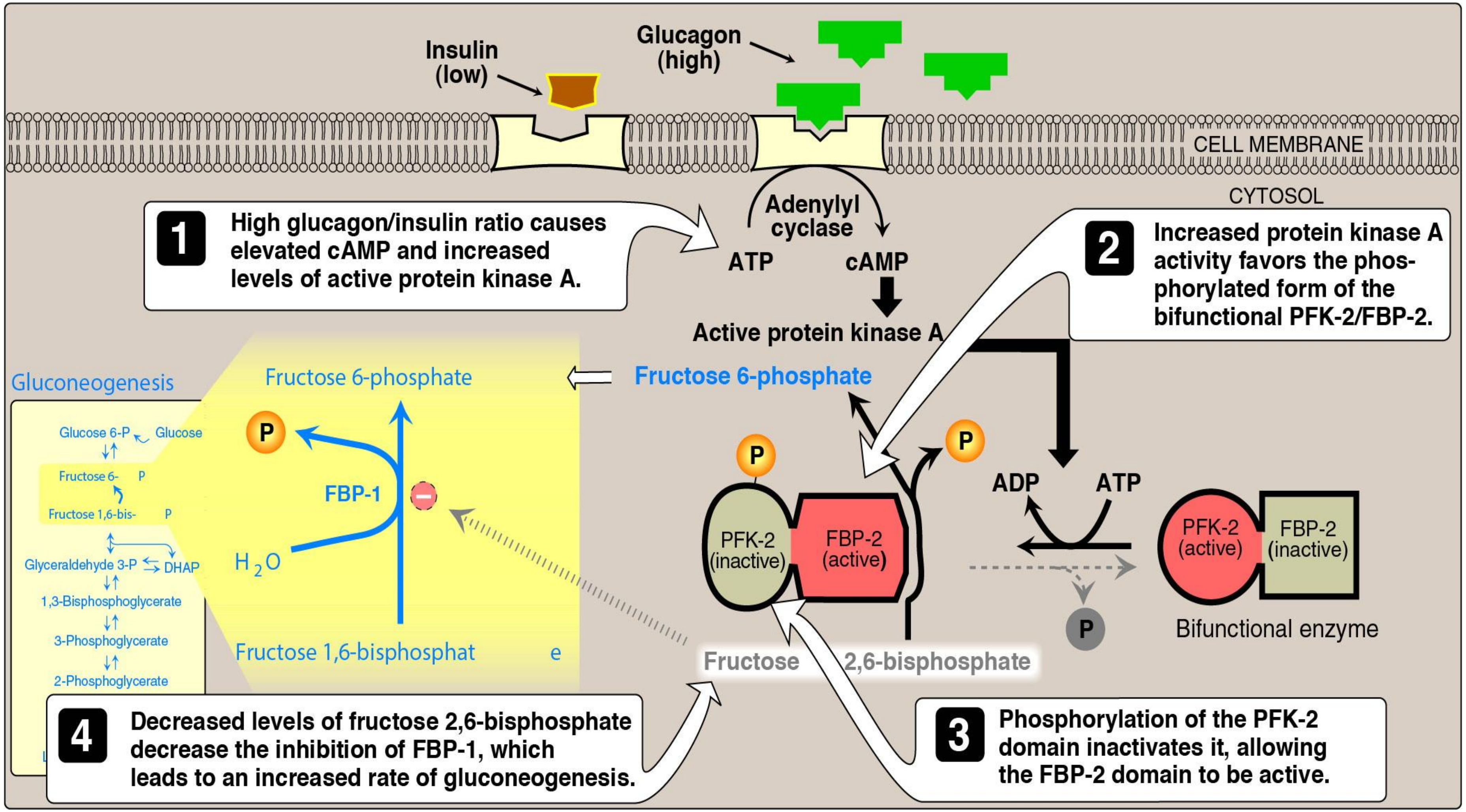
- **Fasting: Glucagon** leads to **phosphorylation** of bifunctional enzyme (via protein kinase A)
  - Causes a conformational shift → Inactivates PFK-2 and activates BPase-2 → **Degrades Fructose 2,6 bisphosphate** → Glycolysis less active; **Gluconeogenesis active**
- **Fed state: Insulin** leads to **dephosphorylation** of bifunctional enzyme
  - Activates PFK-2 and inactivates BPase-2 → Forms Fructose 2,6 bisphosphate → **Glycolysis active**; Gluconeogenesis less active

Summary: Role of bifunctional enzyme

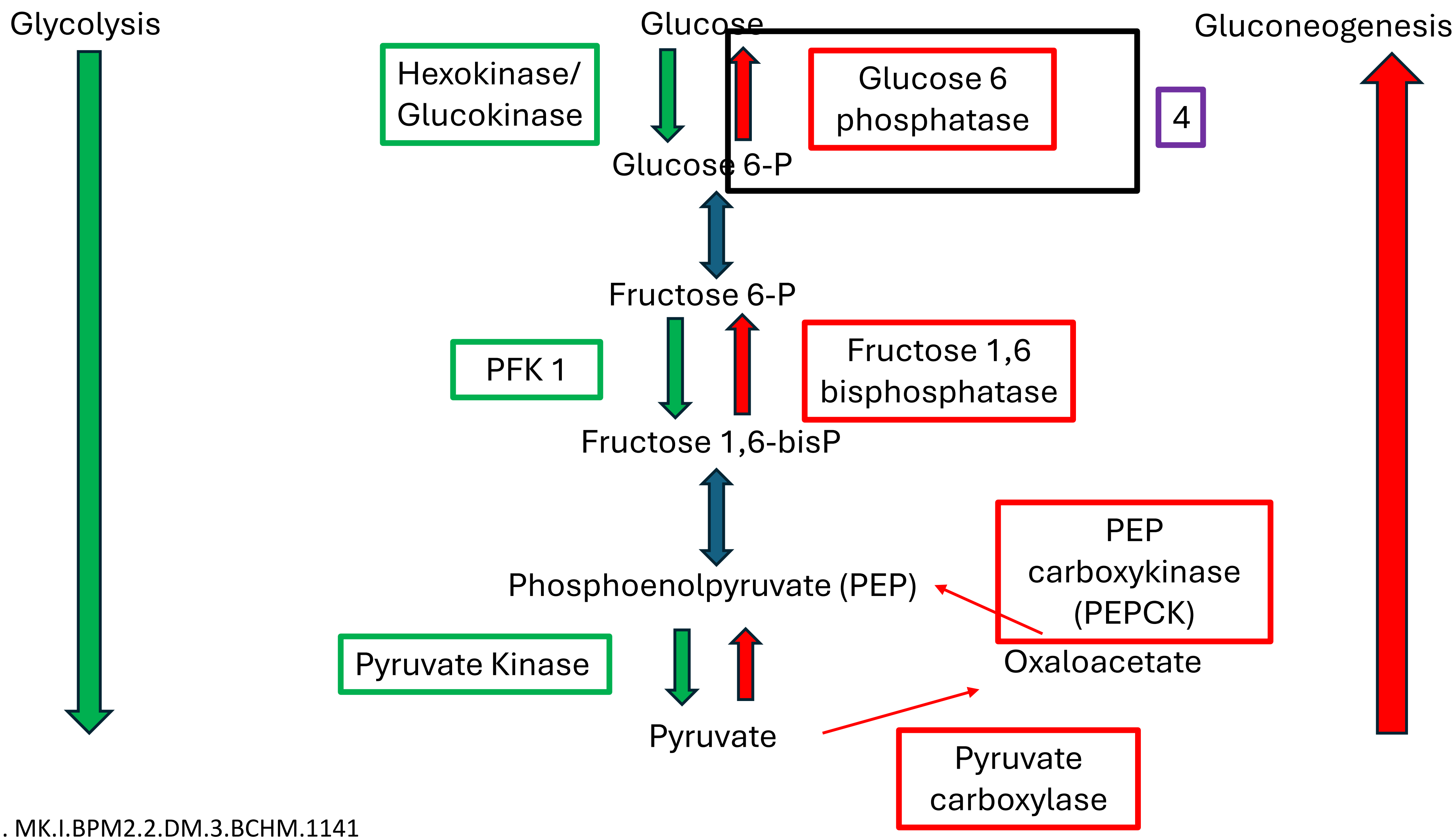
Regulation of Gluconeogenesis vs Glycolysis

	Glucagon/Epinephrine	Insulin
Domain of bifunctional enzyme active	Fructose 2,6 bisphosphatase (FBPase-2)	Phosphofructokinase 2
Effect on Fructose 2,6 bisphosphate levels	Degradation (decreases)	Synthesis (increases)
Effect on glycolysis	Phosphofructokinase-1 **inhibited**.	Phosphofructokinase-1 **stimulated**.
Effect on gluconeogenesis	Fructose 1,6 bisphosphatase **stimulated**	Fructose 1,6 bisphosphatase **inhibited**

Regulation of Fructose 1,6-bisphosphatase



Step 4: Glucose 6-phosphatase



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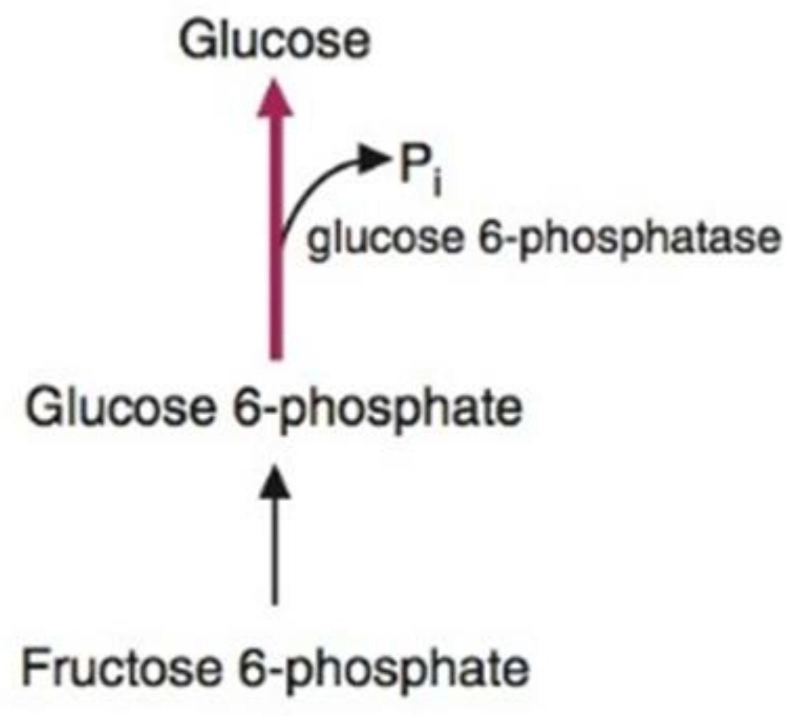
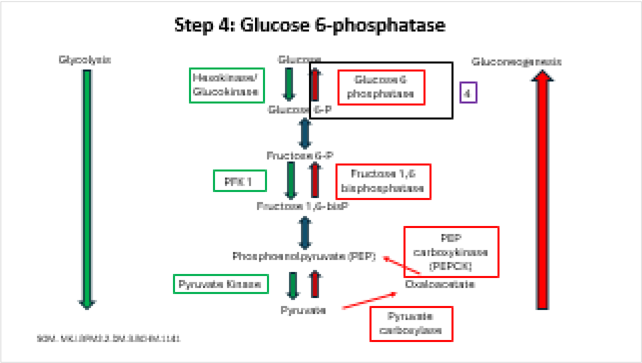
Step 4: Glucose 6-phosphatase: Glucose 6-phosphate to Glucose

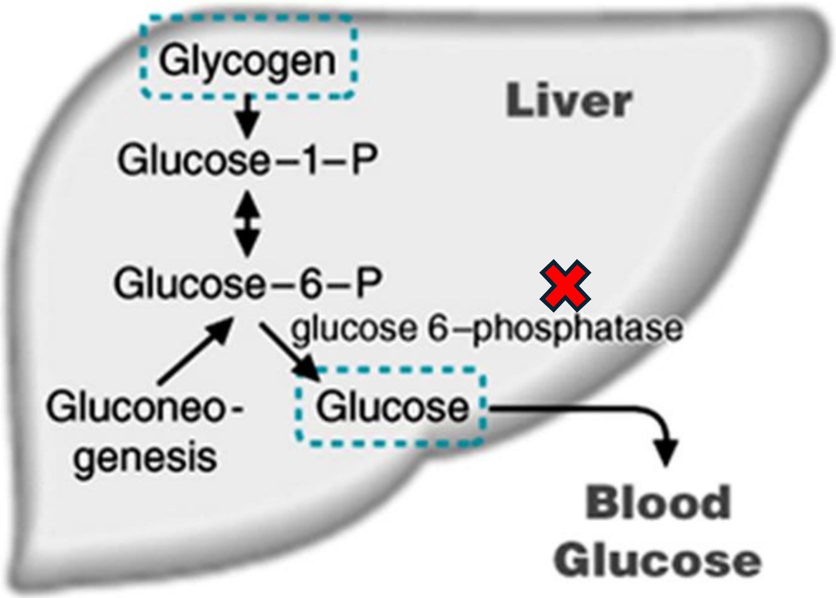


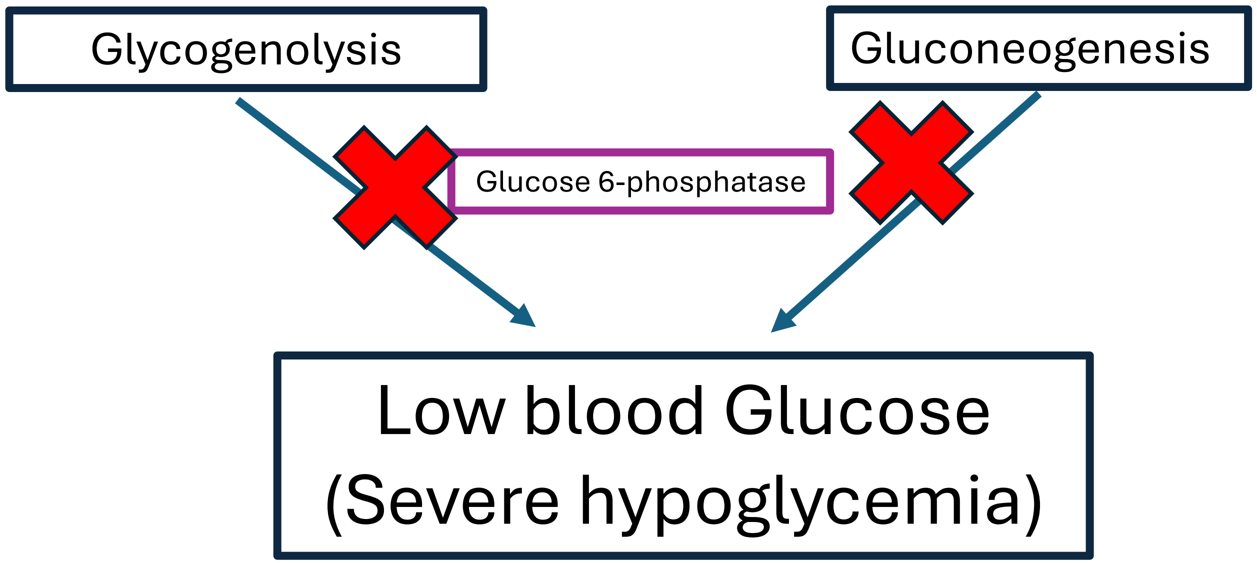
- **Remember**: Glucose 6 phosphatase only located in liver and renal cortex
  - Endoplasmic reticulum
- Required for both gluconeogenesis and glycogen breakdown!!!
- Forms free glucose → Released into blood → Increases plasma glucose level
- Induced (increase gene transcription) during fasting



Von Gierke Disease

(Glucose 6-phosphatase deficiency)





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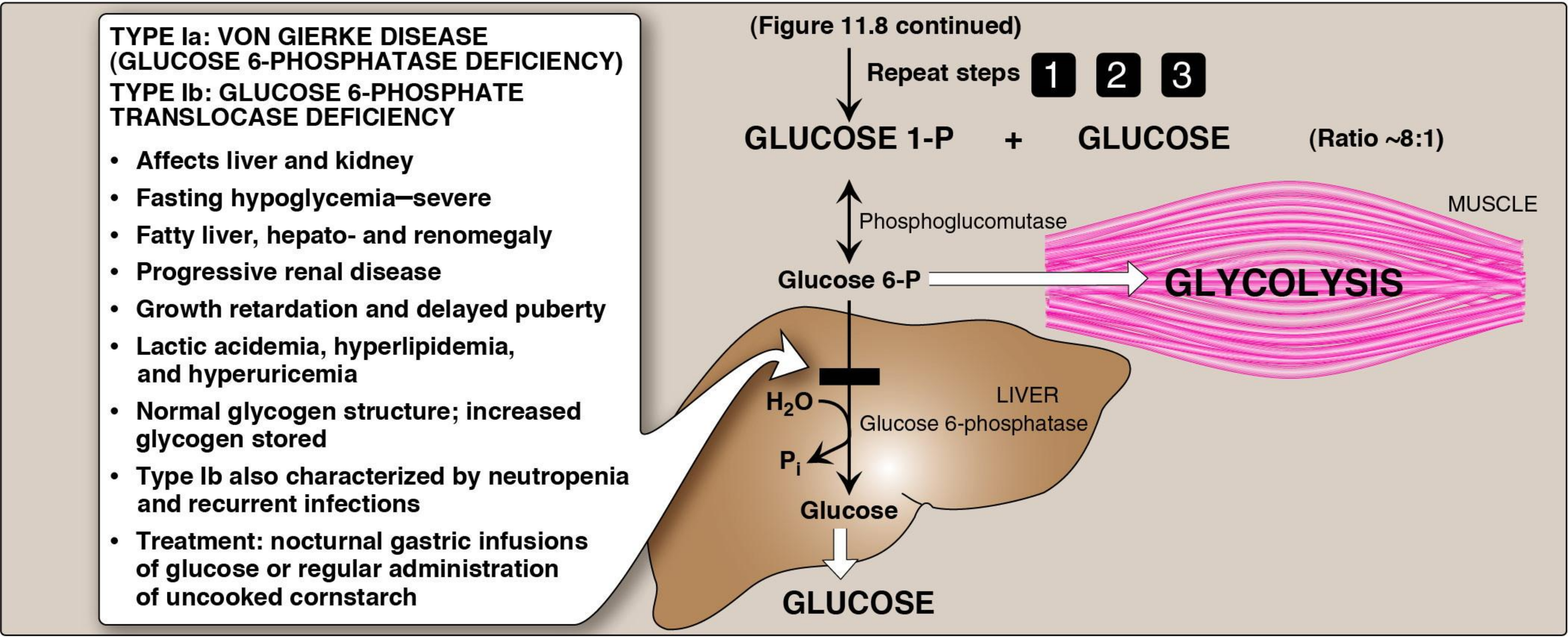
Von Gierke Disease

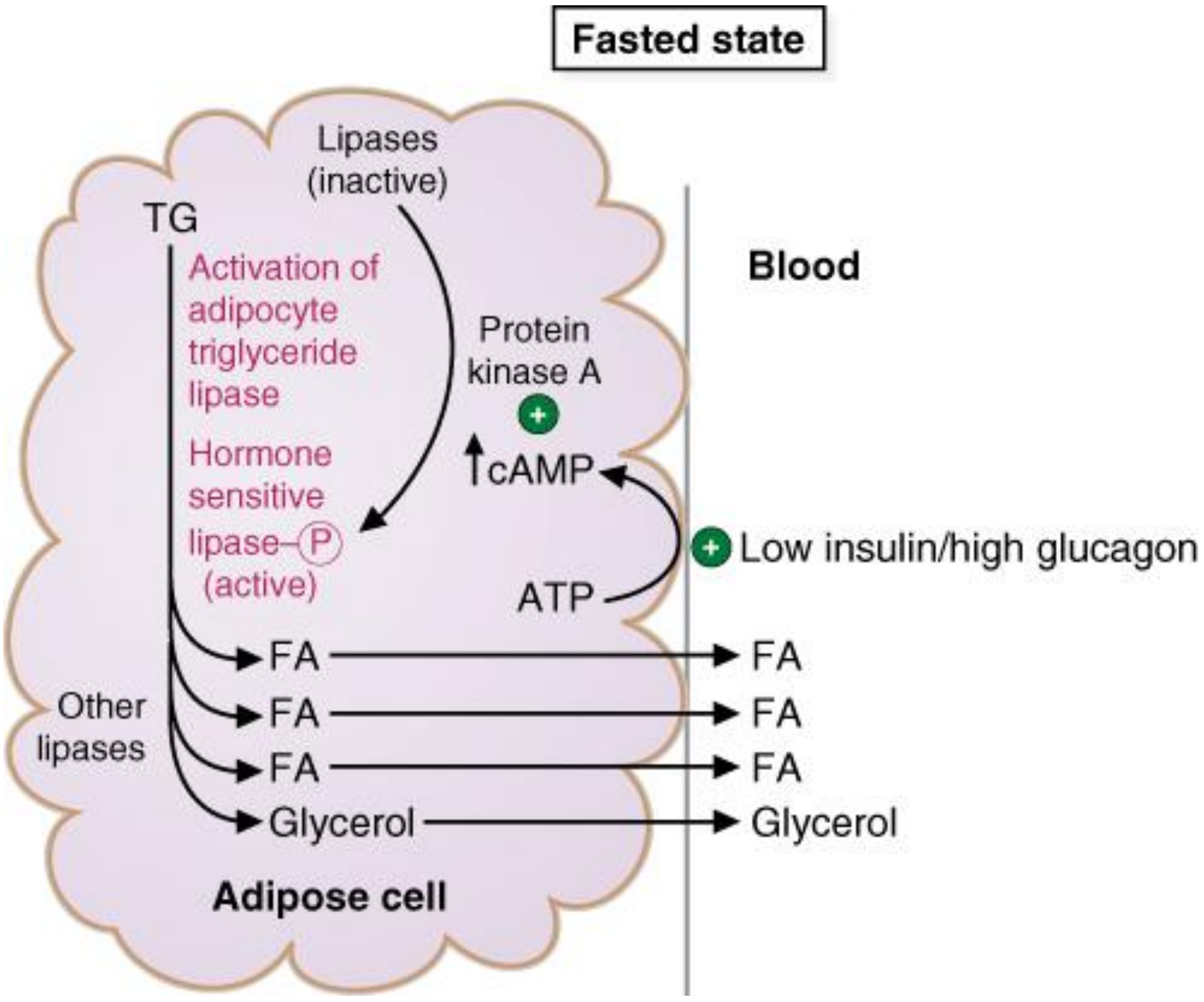
- Glucose 6-phosphatase deficiency
- Most common glycogen storage disorder (autosomal recessive)
- Defect in gluconeogenesis and glycogenolysis → Inability to increase plasma glucose because of defective production of glucose → **Severe hypoglycemia**
- Increased blood lactic acid **(Due to defect in Cori cycle) → Lactic acidosis**
    - Lactate formed in periphery is not converted to glucose due to gluconeogenesis defect)
- Excessive glycogen storage (**Hepatomegaly**)
- Excessive TAG mobilization from adipose tissue → **Lipidemia and ketosis** (ketone body formation)
- Persistent hypoglycemia reduces protein synthesis (chronic low insulin levels) → **Stunted growth**.
- Increased blood lactic acid competes with urate excretion by kidneys → Increased blood uric acid levels → Gout

Von Gierke Disease



TAG degradation (adipose tissue lipolysis)

- TAG degradation (lipolysis) releases fatty acids from adipose tissue
- Fatty acid β-oxidation in liver produces:
  - $FADH_2$ and NADH
  - Acetyl CoA



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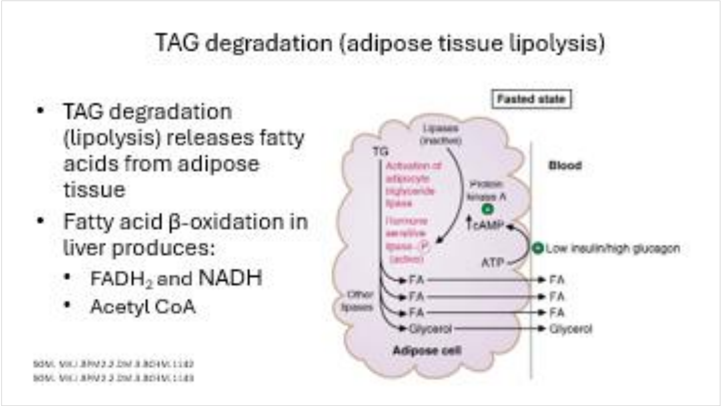
Interrelations between β-oxidation and Gluconeogenesis

- Hepatocytes use **fatty acid β-oxidation** as fuel during **low blood glucose** levels (fasting)
- **Fatty acid β-oxidation** produces:
  - **$FADH_2$ and NADH provide ATP** for gluconeogenesis (fasting/low blood glucose levels)
  - **Acetyl CoA**: Absolute allosteric activator of Pyruvate Carboxylase (gluconeogenesis)
- During fasting, Gluconeogenesis is active (Liver does NOT use glucose).
- NADH also inhibits TCA cycle (Isocitrate dehydrogenase)
- NADH and Acetyl CoA inhibits Pyruvate Dehydrogenase complex



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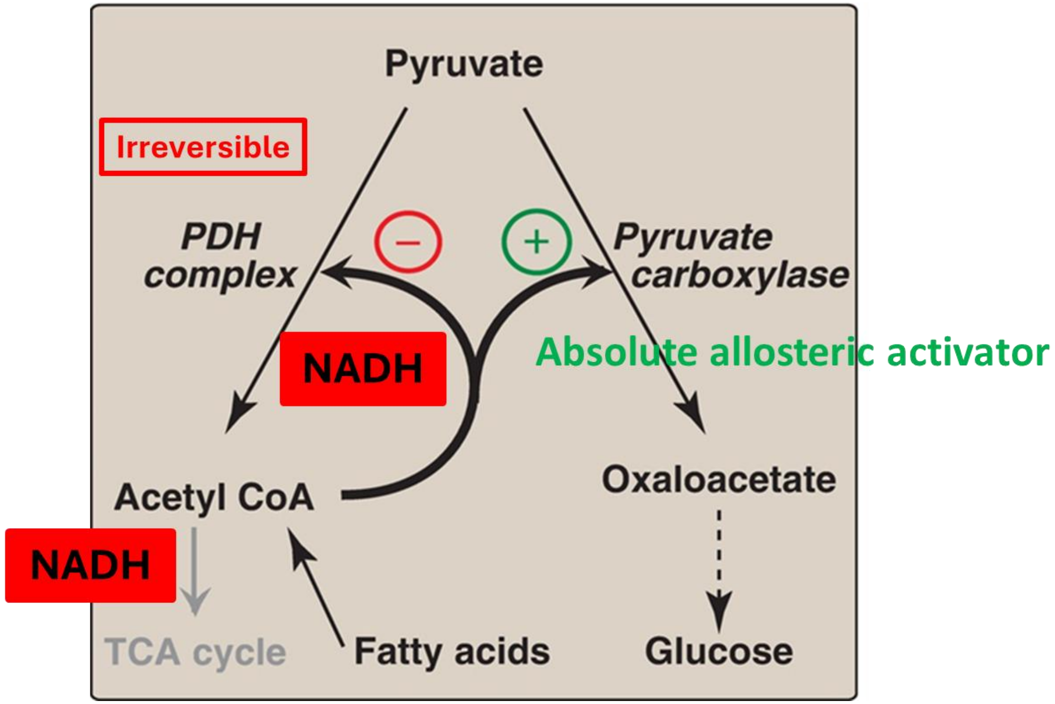
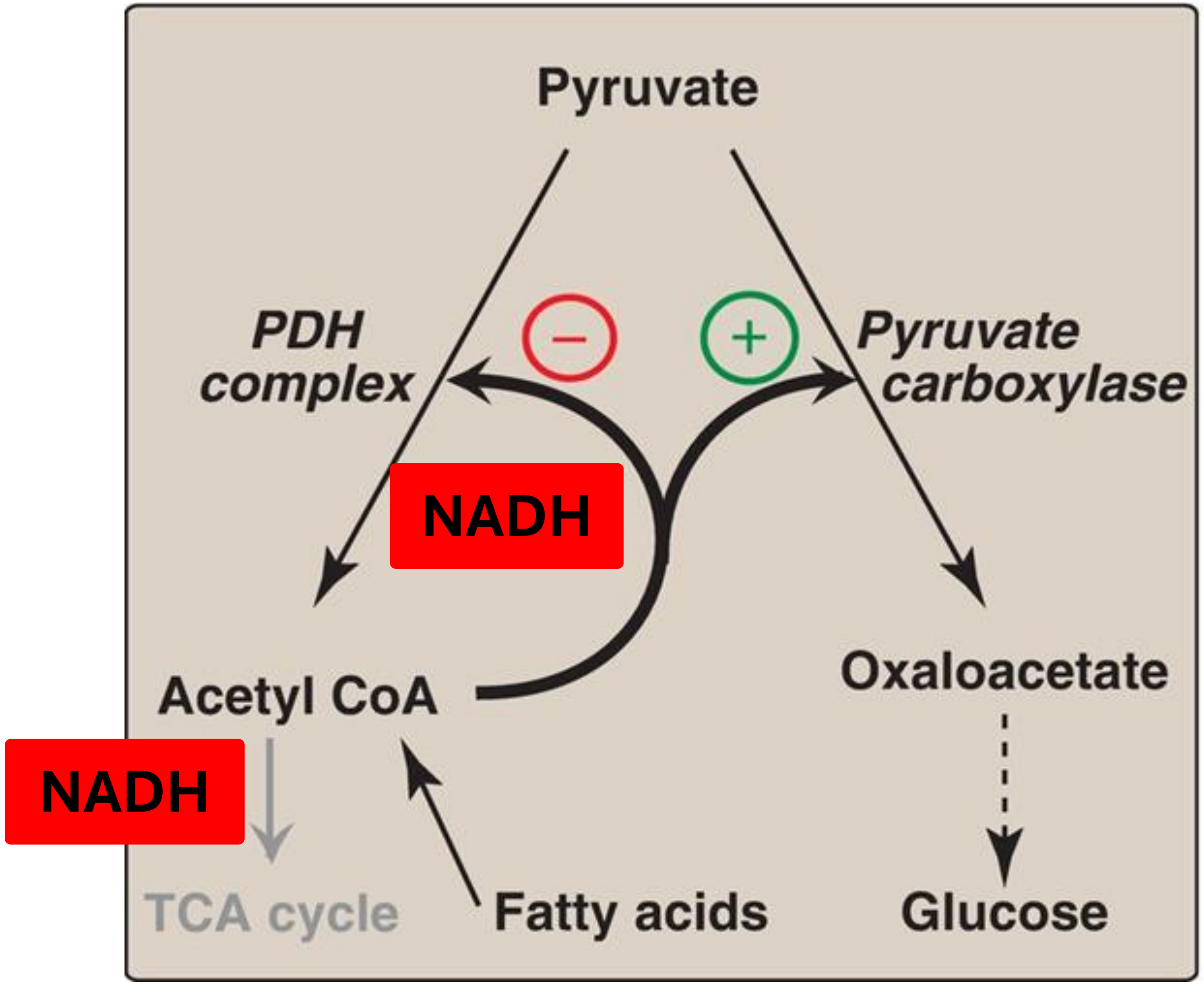
Acetyl CoA and Gluconeogenesis

Acetyl CoA cannot be used to form glucose
because:

- Acetyl CoA formed during fasting cannot form pyruvate because PDH complex is **irreversible**
- Acetyl CoA forms citrate (Citrate synthase in TCA cycle). During fasting TCA cycle is less active.
- The two carbons of Acetyl CoA lost in two decarboxylation steps in TCA cycle
- **Acetyl CoA (formed by β-oxidation of fatty acids): Absolute allosteric activator for pyruvate carboxylase (gluconeogenesis)**



Role of NADH and Acetyl CoA from β-oxidation



β- oxidation

Acetyl CoA

NADH

Absolute activator of **Pyruvate carboxylase (Gluconeogenesis active)**

Activate PDH kinase

Phosphorylation of Pyruvate DH complex **(Inhibits PDH)**

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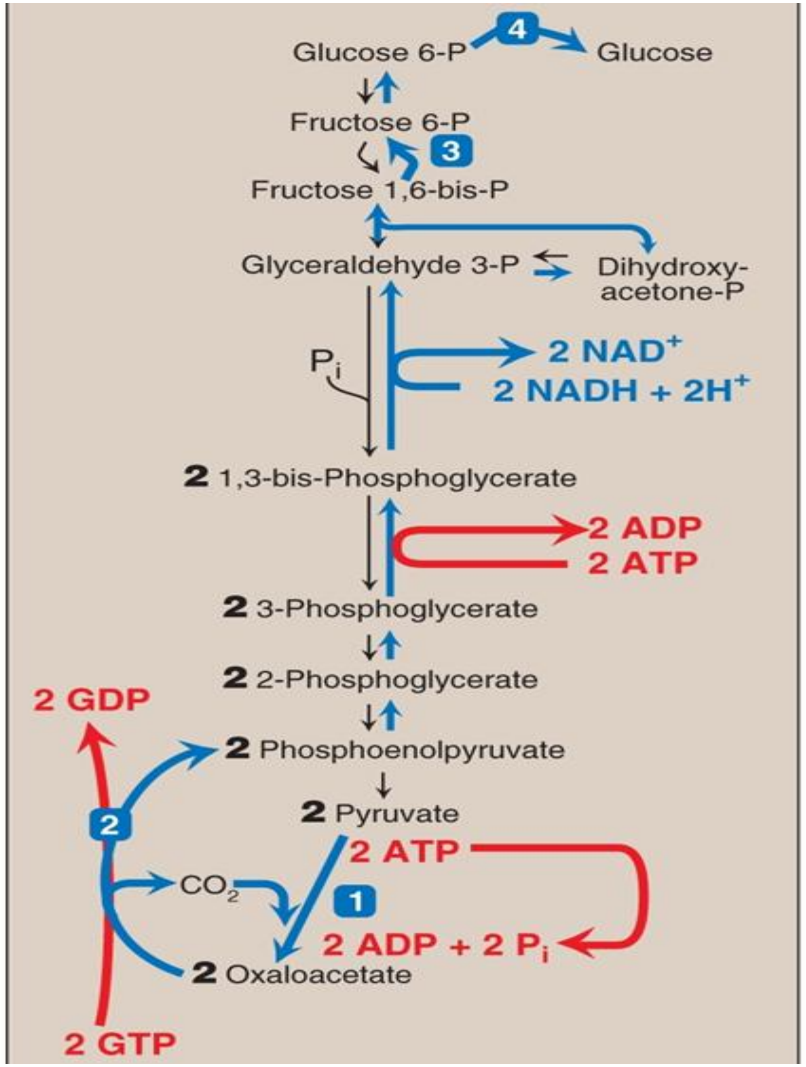
Interrelations between β-oxidation and Gluconeogenesis



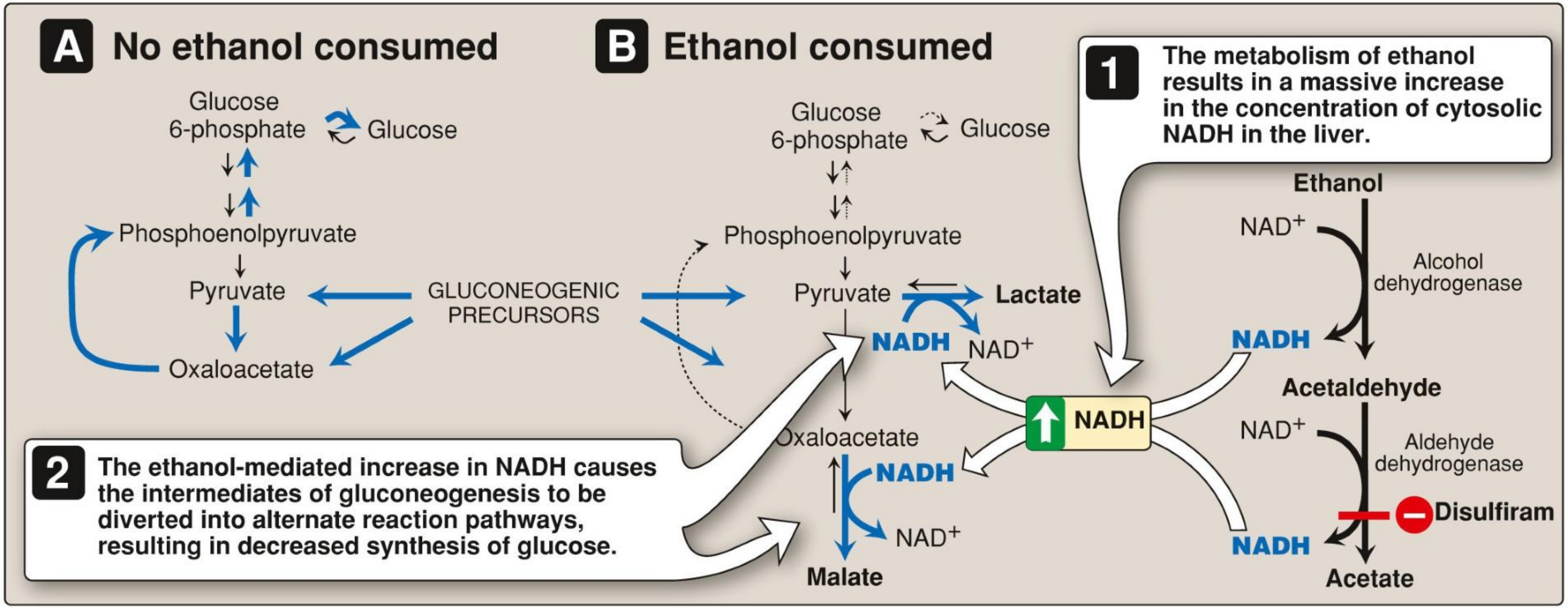
Gluconeogenesis (Anabolic reaction) uses:

- ENERGY (ATP, GTP and NADH)
  - ATP (GTP) and NADH are provided by mitochondrial β-oxidation of fatty acids
- Acetyl CoA: (From β-oxidation) Absolute activator of Pyruvate carboxylase (step 1)
- **Important: Patients with defective fatty acid oxidation will have reduced gluconeogenesis**

Disorders associated with inadequate gluconeogenesis (Hypoglycemia)

- **Type 1 glycogen storage disorder** (Von Gierke disease)
- **Insulinoma**: Insulin inhibits both glycogenolysis and gluconeogenesis.
- Preterm neonates: Delayed expression of enzymes for gluconeogenesis.
- **Addison disease**: Low glucocorticoids/cortisol → Defective gluconeogenesis
- Excess alcohol (**binge alcohol consumption**):
  - Alcohol metabolism forms NADH
  - Converts pyruvate to lactate → Less gluconeogenesis substrates
  - Converts oxaloacetate to malate → Less gluconeogenesis substrates

Binge alcohol consumption



Disorders related to excessive gluconeogenesis (Hyperglycemia)

- **Diabetes Mellitus**: Type 1 (low insulin) and type 2 (insulin resistance).
  - Remember: Normally, insulin inhibits gluconeogenesis
  - Defective inhibition of gluconeogenesis due to low insulin production (Type 1) or insulin resistance (Type 2).
- **Cushing Syndrome:** Excessive cortisol → Induces PEPCK and stimulates muscle protein catabolism (provides amino acids).
- **Hyperthyroidism:** Thyroxine increases hepatocyte sensitivity to epinephrine and cortisol. (also inhibits insulin secretion).
- **Pheochromocytoma:** Chromaffin cell tumor secretes norepinephrine and epinephrine → Stimulate gluconeogenesis
- **Glucagonoma**: Pancreatic tumor secreting excessive glucagon → Stimulates gluconeogenesis

Summary

Fasting	**Low blood glucose**, **low** insulin/glucagon ratio. Fasting, fight and flight situations (glucagon, epinephrine, cortisol blood levels increase).
Pathways active in liver	**Gluconeogenesis** and **fatty acid oxidation**, **glycogenolysis** mainly in hepatocytes.
Irreversible steps of gluconeogenesis	**Pyruvate carboxylase (**mitochondria) contains **biotin; Acetyl CoA** is absolute activator; Pyruvate → Oxaloacetate **PEP carboxykinase (PEPCK) –** cytosol; Needs GTP; Oxaloacetate → Phosphoenol pyruvate (PEP) **Fructose 1,6-bisphosphatase (**cytosol); Allosteric inhibitor – Fructose 2,6BisP; Fructose 1,6-bisphosphate → Fructose 6-phosphate **Glucose 6-phosphatase** in ER; Glucose 6-phosphate → Glucose
Bifunctional enzyme	Phosphorylation by PKA following **glucagon** action (cAMP-messenger system): inactive PFK-2 and active BPase-2 **inhibit glycolysis** and **stimulates gluconeogenesis**. **Low levels of Fructose 2,6-bisP → Activates gluconeogenesis.**
Allosteric regulations	**Acetyl CoA** from fatty acid degradation (**b-oxidation)** allosterically **activates pyruvate carboxylase.** **PDH complex** is product **inhibited** by acetyl CoA and NADH which activate **PDH kinase**. **NADH** allosterically **inhibits** the **TCA cycle mainly** at isocitrate DH. **AMP** or **fructose 2,6-bisP** allosterically inhibit **fructose 1,6-bisphosphatase**.
Phosphorylations	**Glucagon : Phosphorylation** of the **bifunctional enzyme** and degradation of F2,6-bisP → Activates gluconeogenesis **Phosphorylation** of the **PDH complex** by PDH kinase that inhibits the PDH complex. Inhibition of glycolysis by **phosphorylation of pyruvate kinase** and PEP used for gluconeogenesis.
Hormones	**Glucagon (fasting) induces** glucose 6-phosphatase, fructose 1,6-bisphosphatase and PEPCK. **Cortisol induces** PEPCK and is needed for long-term gluconeogenesis. **Epinephrine (stress)** activates gluconeogenesis **Insulin inhibits gluconeogenesis**
Substrates for gluconeogenesis	**Alanine** and **lactate** from the blood are used to form pyruvate. **Glucogenic amino acids: Alanine, Glutamine, Glutamate** **Glycerol** from the adipose tissue TAG